

COORDINATION CHEMISTRY

- 1 Compound $\text{CoCl}_3 \cdot 4\text{NH}_3$ is observed in two different colours green and violet but distinct properties they are—
 (1) Optical isomers
 (2) geometrical isomers
 (3) Ionisation isomers
 (4) coordination isomers
- 2 Which among the following statements is wrong—
 (1) Primary valencies are ionisable
 (2) Secondary valencies are satisfied by the ligands
 (3) Primary valencies are equal to coordination number of central metal ion
 (4) Primary valencies are satisfied only by the anions
- 3 The complex having square planar shape is—
 (1) $[\text{Ni}(\text{CN})_4]^{4-}$ (2) $[\text{Ni}(\text{CO})_4]$
 (3) $[\text{Cd}(\text{CN})_4]^{2-}$ (4) $[\text{PtCl}_4]^{2-}$
- 4 Which of the following will give two mole of AgCl precipitated per mole of the compound with excess AgNO_3 ?
 (1) $\text{PtCl}_4 \cdot 2\text{HCl}$ (2) $\text{PtCl}_2 \cdot 2\text{NH}_3$
 (3) $\text{PtCl}_2 \cdot 2\text{en}$ (4) $\text{PtCl}_4 \cdot 5\text{NH}_3$
- 5 Which ligand on coordination with central metal ion, form maximum number of ring structures?
 (1) Triphenyl phosphine (PPh_3)
 (2) diethylene triamine (dien)
 (3) Triethylenetetraamine (trien)
 (4) Acetylacetonate (acac)
- 6 The heteroleptic complex is—
 (1) Cis-platin
 (2) Brown ring complex
 (3) Zeise salt
 (4) All
- 7 The IUPAC name of the complex $[\text{Co}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_3]_2(\text{SO}_4)_3$ is—
 (1) Bis (triethylenediamine) cobalt (III) disulphate
 (2) Tris (ethylenediamine) cobalt (III) sulphate
 (3) Triethylene diamine dicobalt (II) sulphate
 (4) Tris (ethylenediamine) cobalt (II) trisulphate

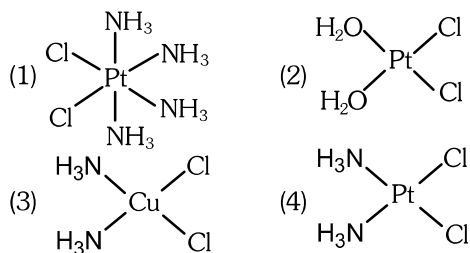
- 8 Tetraammineaquachloridocobalt (III) chloride can exhibit—

- (1) Geometrical and optical isomerism
 (2) Geometrical and ionisation isomerism
 (3) Optical and linkage isomerism
 (4) Only geometrical isomerism

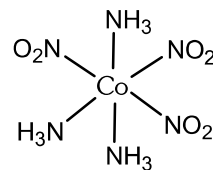
- 9 Mercury tetrathiocyanato Cobaltate (III) is—

- (1) $[\text{Hg}(\text{SCN})_2][\text{Co}(\text{SCN})_2]$
 (2) $\text{Co}[\text{Hg}(\text{CN})_4]$
 (3) $\text{Hg}[\text{Co}(\text{NCS})_4]$
 (4) $\text{Hg}[\text{Co}(\text{SCN})_4]$

- 10 Complex which is used as anticancer drug is—



- 11 The complex



is—

- (1) laevo-form (2) Facial-form
 (3) cis-form (4) Meridional-form

- 12 Which type of isomerism is exhibited by $\text{K}[\text{Cr}(\text{H}_2\text{O})_2(\text{C}_2\text{O}_4)_2]$?

- (1) Ionisation (2) Hydrate
 (3) Geometrical (4) All

- 13 The spin only magnetic moment of $[\text{MnBr}_4]^{2-}$ is 5.9 B.M. The correct statement about the complex is —

- (1) It has tetrahedral geometry
 (2) Configuration is $t_2g^5 e_g^0$
 (3) Configuration is $t_2g^2 e_g^3$
 (4) It is low spin complex

- 14** Crystal field splitting in octahedral complex causes—
- (1) Stabilisation of e_g energy set
 - (2) Lowering of energy of d_{xy} , d_{yz} , and d_{xz} by $2/5\Delta_0$
 - (3) Lowering of energy of $d_{x^2-y^2}$, d_{z^2} by $3/5\Delta_0$
 - (4) increase in energy of t_{2g} set
- 15** In d^4 configuration when $\Delta_0 < P$ fourth electron enters in —
- (1) d_{xy} orbital
 - (2) t_{2g} energy set
 - (3) d_{xz} orbital
 - (4) e_g energy set
- 16** For the same metal, and same ligands, the relation in between Δ_t and Δ_0 is—
- (1) $\Delta_t = \frac{9}{4}\Delta_0$
 - (2) $\Delta_t = \frac{3}{5}\Delta_0$
 - (3) $\Delta_t = 0.45\Delta_0$
 - (4) $\Delta_t = 0.6\Delta_0$
- 17** In zeise salt, Pt to C_2H_4 bonding creates a synergic effect what is true regarding these two species—
- (1) Pt is π -acceptor as well as σ -donor
 - (2) Pt is σ -acceptor and π -donor
 - (3) Pt is σ -donor and σ -acceptor
 - (4) C_2H_4 is π -donor and π -acceptor
- 18** Pick out the false statement about analytical applications of complex—
- (1) Brown ring complex is formed during detection of NO_3^- by acidified $FeSO_4$.
 - (2) DMG is used to detect Ni^{2+} ion in the solution
 - (3) EDTA is used to estimate hardness of water
 - (4) $K_3[Fe(CN)_6]$ is used to detect Fe^{3+} ion
- 19** Pick out the incorrect match among the following
- (1) Vitamin B_{12} – the antipernicious anaemia factor
 - (2) $[RhCl(PPh_3)_3]$ – Hydrogenation of alkenes
 - (3) $[Ag(CN)_2]^-$ – Photography
 - (4) D- penicillamine & desferrioxime-B is used to remove excess of Cu and Fe from plant/animal system.